INCH-POUND

MIL-DTL-32113 (OS)
09 September 2003

DETAIL SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, JAU-74/A

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document gives the requirements for the manufacture, assembly, and packaging of the Initiator, Cartridge Actuated, JAU-74/A and the verification methods upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in section 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1. Specification, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-C-83124	Cartridge Actuated/Propellant Actuated Devices, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-810	Environmental Engineering Considerations And Laboratory Tests
MIL-STD-331	Fuze And Fuze Components Environmental And Performance Tests For
DOD-STD-2101	Classification of Characteristics

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Bldg. 4D Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

INSTRUCTION

Naval Sea Systems Command (Cage Code 53711)

8010.5	Insensitive Munitions Program Planning and Execution
8020.8	Department of Defense Ammunition and Explosive Hazard Classification Procedures

Naval Air Systems Command (Cage Code 30003)

4200.25	Engineering Reviews of New Sources for Procurement Repair or Overhaul of Replenishment Critical Safety Items

DRAWINGS

Naval Air Systems Command (Cage Code 30003)

MBEU148144-1	Cartridge – Catapult Initiator (JAU-74/A)
MBEU148480-1	Chamber – Assy. Testing
3205AS284	JAU-74/A Initiator, Shock/Vibration Test Fixture

(Application for copies should addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, MD 20640-5035.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents, which are DOD adopted, are

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those listed in the issue of DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (See 6.2).

American Society for Testing Materials

ASTM E 1742	Standard Practice for Radiographic Examination

(Application for copies should be addressed to the American Society for Testing Materials, 100 Barr Harbor Dr., West Conshohocken, PA 19428)

Society Of Automotive Engineers (SAE)

SAE-AS5604	Compass, Magnetic, Pilot's Standby

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

United Kingdom Royal Ordnance Defence Specifications

AIR 2178	Specification for manufacture and quality assurance of Cap, Percussion, QV 5AF (VH2 Composition filling)
100 454-01/1/WE	Composition SR 371C (Magnesium Powder)

(Application for copies should be addressed to the Technical Document Centre, BAE Systems, RO Defence, Glascoed, Usk, Gwent, NP15 1XL)

United Kingdom Royal Ordnance Bishopton Specifications

BS-T-480-SO5-20129-1	Propellant, K.U.

(Application for copies should be addressed to the BAE Systems, RO Defence, Station Road, Bishopton, Renfrewshire, PA7 5NJ)

United Kingdom Ministry of Defence Directorate of Standardization

13-167 (G40)	Gunpowder, Types: G7, G12, G40 and Mealed

(Application for copies should be addressed to Kentigern House, 65 Brown Street, Glasgow, G2 8EX)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a sample of initiators shall be subjected to first article inspection in accordance with 4.1.1.

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3.2 Serial number. Each initiator shall be identified with a unique serial number starting at 001 and proceeding upward in each lot.

3.3 Flight critical parts. The initiator is a flight critical part and shall be identified and procured as such in accordance with NAVAIRINST 4200.25.

3.3.1 Product baseline. The manufacturer shall maintain a complete product baseline list, specifying each item used in the initiator, which shall contain for each item: description, part number, or identifying number, revision letter and material and/or applicable specification. This list and subsequent revisions shall be submitted for technical review and shall be approved in writing by the Naval Surface Warfare Center, Indian Head, MD 20640-5035.

3.4 Design and construction.

3.4.1 General. The initiator shall be manufactured in accordance with Drawing 30003-MBEU148144-1, meet all requirements listed herein, and the requirements of MIL-C-83124 at any given time.

3.4.2 Primary components. For the purpose of this specification, the KU propellant, magnesium powder, gunpowder, and QV5AF primer are defined as primary components. Only one lot of each primary component shall be used in an initiator lot, however, one primary component lot may be used in more than one lot.

3.4.3 Gunpowder. Gunpowder used in the initiator shall meet the requirements of CS 5110 – G40.

3.4.4 KU propellant. KU propellant used in the initiator shall meet the requirements of CS 5470.

3.4.5 Magnesium powder. Magnesium powder used in the initiator shall meet the requirements of CS 5363.

3.4.6 QV5AF primer. The primers shall meet the requirements of AIR 2178, and shall have been manufactured within the last 24 months or re-certified within the last 12 months.

3.5 Performance conditions.

3.5.1 Visual inspection. Initiators shall have proper identification markings and shall be free of defects.

3.5.2 Radiographic examination. The initiator shall be free of defects or foreign material.

3.5.3 Leakage. Each initiator shall meet the requirements of this section.

3.5.3.1 Production installs. All initiators built for fleet installation shall exhibit a leakage rate not greater than 7.6×10^{-7} cm^3/s of helium at a pressure differential of 1 ± 0.1 atmosphere.

3.5.3.2 Non-Production installs. Each initiator not built for fleet installation when subjected to LAT, Design Certification and First Article Testing shall not exhibit a leakage rate greater than 1×10^{-5} cm^3/s of air at a pressure differential of 1 ± 0.1 atmosphere.

3.5.4 Residual magnetism. The initiator shall be free of residual magnetism.

3.5.5 Firing mechanism. Each assembled firing mechanism shall be capable of meeting the requirements given below. The number of firing mechanisms selected from each production lot to verify this shall be in

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accordance with 4.1.4.1. All mechanisms that are subjected to these tests shall be discarded or certified acceptable by Quality Assurance before re-assembly into an initiator body.

3.5.5.1 Free movement. The free movement (without O-ring installed), proper alignment, and proper protrusion of the firing pin in the firing pin housing shall be verified for each initiator during assembly.

3.5.5.2 Non-actuation. The firing mechanisms shall not actuate when subjected to a pressure of 400 +0/-10 psig for 1 minute after being applied at an approximate rate of 200 psig/sec.

3.5.5.3 Actuation. The same firing mechanism previously subjected to the non-actuation shall actuate between 400 and 625 psig when subjected to a pressure in 900 psig at a rate of 10,000 to 50,000 psig/sec.

3.6 Environmental conditions.

3.6.1 Low temperature conditioning. The initiator shall meet the ballistic conditions of section 3.7 when functioned after being stabilized at –65 ± 5 °F for a minimum of twelve hours.

3.6.2 Normal temperature conditioning. The initiator shall meet the ballistic conditions of section 3.7 when functioned after being stabilized at 70 ± 5 °F for a minimum of twelve hours.

3.6.3 High temperature conditioning. The initiator shall meet the ballistic conditions of section 3.7 when functioned after being stabilized at 200 ± 5 °F for a minimum of twelve hours.

3.6.4 Six-foot drop. The initiator shall not initiate or incur any defects or degradation, and shall meet all performance conditions after being dropped from a height of 6 feet, +3/-0 inches onto a two-inch thick steel plate imbedded in concrete. The initiator shall also meet the ballistic conditions of section 3.7 when functioned

3.6.5 Forty-foot drop. The initiator shall not initiate or incur any internal/external defects or degradation that makes the initiator unsafe for disposal when dropped from a height of 40 feet, +3/-0 inches onto a two inch thick steel plate imbedded in concrete.

3.6.6 Salt fog. The initiator shall meet performance conditions after being subjected to the salt fog test, MIL-STD-810, Method 509.2 for 168 hours. The initiator shall also meet the ballistic conditions of section 3.7 when functioned.

3.6.7 High temperature. The initiator shall not have any defects or degradation when exposed to a temperature of 200 ± 5 °F for 50 hours. The initiator shall also meet the ballistic conditions of section 3.7 when functioned.

3.6.8 Iced condition. The initiator shall not incur any defect or degradation when submitted to the iced condition test and ballistically tested at –80 ± 5 °F.

3.6.9 Dust. The initiator shall not incur damage or degradation and shall meet performance conditions after being subjected to the blowing dust (small particle) test, MIL-STD-810, Method 510.3. The initiator shall also meet the ballistic conditions of section 3.7 when functioned.

3.6.10 Shock. The initiator shall not fire or incur any defects or degradation as a result of the 20g sawtooth pulse of 11 millisecond duration and shall meet all performance conditions. The initiator shall also meet the ballistic conditions of section 3.7 when functioned.

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3.6.11 Temperature-shock/humidity/altitude (TSHA) cycling. The initiator shall not fire or incur defects or degradation as a result of the 28 day TSHA cycling test and shall meet all performance conditions. The initiator shall also meet the ballistic conditions of section 3.7 when functioned.

3.6.12 Low temperature exposure. The initiator shall not fire or incur defects or degradation when stabilized at –60 ± 5 °F for 72 hours and shall meet all performance conditions. The initiator shall also meet the ballistic conditions of section 3.7 when functioned.

3.6.13 Vibration. The initiator shall not fire or incur defects or degradation as a result of the non-gunfire and gunfire profiles and shall meet all performance conditions. The initiator shall also meet the ballistic conditions of section 3.7 when functioned.

3.6.14 Cook off. The initiator shall be tested to determine the highest temperature at which no cartridge cooks off within a one-hour period. Cook-off temperature shall not be less than 225 °F.

3.6.15 High temperature exposure. The initiator shall be tested to determine the highest temperature to which the initiator may be exposed for a period of 12 hours without affecting ballistic conditions. The high temperature exposure limit determined shall not be less than 200 °F.

3.7 Ballistic conditions. The initiator when ballistically tested in conjunction with the CCU-146/A Impulse Cartridge shall meet the requirements shown in Table I.

TABLE I. Ballistic parameters

Temperature (^{o}F)	Ignition delay (ms)	Maximum Pressure (psig)	Time to Maximum pressure (ms)	Time to ½ Maximum pressure (ms)	Actuation Pressure (psig)
-65	15 (max)	1120-1380	75-114	230-300	400-625
70	15 (max)	1220–1500	61-93	210-270	400-625
200	15 (max)	1290-1590	53-81	190-260	400-625

3.7.1 Integrity. No initiator, its subassemblies, or individual part shall fragment, rupture, and fail causing a marginal condition.

3.7.2 Misfires. There shall be no misfires.

3.7.2.1 Re-tests. There shall be no re-tests

3.7.2.2 Test failure. Any initiator that does not meet performance conditions of paragraphs 3.5 through 3.7 during any phase of inspection shall be considered to have failed.

3.7.2.3 Acceptance criteria. Initiators shall meet the requirements of 3.5 through 3.7. The acceptable number of defects is 0 and the reject number is 1.

3.8 Insensitive munitions. The initiator shall be certified as an insensitive munition to external stimuli.

3.9 Explosive hazard classification. The initiator in its shipping condition shall have a hazard classification of 1.4 with a compatibility code of “C”.

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3.10 Workmanship. Initiators shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Initiators shall have proper identification markings and shall be free of defects.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1)
b. Design certification inspection (4.1.2)
c. Production inspection (see 4.1.3)
d. Lot acceptance inspection (see 4.1.4)

4.1.1 First article inspection. Unless otherwise specified in the contract or purchase order a first article sample of 69 initiators shall be prepared. This sample, manufactured using the same methods and procedures proposed for the production lot, is for the purpose of determining that the contractor, prior to the starting production, is capable of producing items, which comply with the technical requirements of the contract. Sixty-six of the initiators shall be expended in the tests listed in Table II and three shall be retained as spares/investigative purposes. A first article sample acceptable for environmental and ballistic conditions shall be defined as a sample which has met the requirements of Table IV of this specification. The contractor shall deliver the initiators to the activity designated by the contracting agency for test. First article acceptance shall consist of the tests shown in Table II in the order specified. Any quantity production prior to notification by the contracting agency of the first article sample acceptability shall be at the contractor's risk. The first article initiators will not be applied as a part of the quantity specified by the contract.

4.1.2 Design certification inspection. Unless otherwise specified in the contract or purchase order a design certification sample of 134 initiators shall be prepared. This sample, manufactured using the same methods and procedures proposed for the production lot, is for the purpose of determining that the manufacturer, prior to the starting production, complies with the technical requirements of the contract. One hundred and twenty six of the initiators shall be expended in the tests listed in Table III and two shall be retained as spares/investigative purposes. A design certification sample acceptable for environmental and ballistic conditions shall be defined as a sample that has met the requirements of Table IV of this specification. The contractor shall deliver the initiators to the activity designated by the contracting agency for test. Manufacturer design certification shall consist of the tests shown in Table III in the order specified. Any quantity production prior to notification by the contracting agency of the sample acceptability shall be at the contractor's risk. The certification initiators will not be applied as a part of the quantity specified by the contract.

4.1.3 Production inspection. All initiators manufactured under a contract shall be inspected and screened for defects. Initiators failing to meet the requirements that are listed in Table IV shall be rejected and removed from the lot.

4.1.3.1 Classification of characteristics. The characteristics verified by tests and examinations herein are classified as Critical, Major, or Minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of the test or examination. Tests and examinations that are not annotated with a classification code are classified minor.

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TABLE II. First article inspection

Test/Inspection	Requirement Paragraph	Procedure Paragraph	6	6	6	36	27	18	9	3	3	6	3
Baseline – Non-Destructive													
Visual Inspection	3.5.1	4.5.1	6	6	6	36				3	3	6	3
Radiographic Examination	3.5.2	4.5.2	6	6	6	36				3	3	6	3
Leakage	3.5.3	4.5.3	6	6	6	36				3	3	6	3
Residual magnetism	3.5.4	4.5.4	6	6	6	36				3	3	6	3
Baseline – Destructive													
Low Temp. Conditioning	3.6.1	4.6.1	6										
Normal Temp. Conditioning	3.6.2	4.6.2		6									
High Temp. Conditioning	3.6.3	4.6.3			6								
Environmental Tests													
Sequential Program													
20 G Shock	3.6.10	4.6.10				36							
- Visual/X-ray/Leak	3.5.1, .2, .3	4.5.1,.2,.3				36							
TSHA Cycling	3.6.11	4.6.11					27						
- Visual/X-ray/Leak	3.5.1, .2, .3	4.5.1,.2,.3					27						
Low Temp. (-60°F)	3.6.12	4.6.12						18					
- Visual/X-ray/Leak	3.5.1, .2, .3	4.5.1,.2,.3						18					
Vibration (Non-Gunfire)	3.6.13.1	4.6.13.1							9				
- Visual/X-ray/Leak	3.5.1, .2, .3	4.5.1,.2,.3							9				
Vibration (Gunfire)	3.6.13.3	4.6.13.3							9				
- Visual/X-ray/Leak	3.5.1, .2, .3	4.5.1,.2,.3							9				
Safety Tests													
Six Foot Drop	3.6.4	4.6.4								3			
- Visual/X-ray/Leak	3.5.1, .2, .3	4.5.1,.2,.3								3			
Forty Foot Drop	3.6.5	4.6.5									3		
- Visual/X-ray	3.5.1, .2	4.5.1,.2									3		
Firing Mechanism Test	3.5.2, .3	4.5.2, .3										6	
Performance Tests													
Function at -65°F	3.7	4.7	6			3	3	3	3				
Function at 70°F	3.7	4.7		6		3	3	3	3				
Function at 200°F	3.7	4.7			6	3	3	3	3				
Spare Units	-	-											3

TABLE III. Design certification inspection

Test/Inspection	Reqmt. Para.	Procedure Para.	I	II	III	IV	V	VI	VII	VIII-A	VIII-B	VIII-C	VIII-D	IX	X	XI	XII	XIII	XIV
			12	12	12	12	3	3	6	36	27	18	9	3	3	6	12	12	2
Baseline – Non-Destructive																			
Visual Inspection	3.5.1	4.5.1	12	12	12	12	3	3	6	36				3	3	6	12	12	2
Radiographic Examination	3.5.2	4.5.2	12	12	12	12	3	3	6	36				3	3	6	12	12	2
Leakage	3.5.3	4.5.3	12	12	12	12	3	3	6	36				3	3	6	12	12	2
Residual Magnetism	3.5.4	4.5.4	12	12	12	12	3	3	6	36				3	3	6	12	12	2
Baseline – Destructive																			
Low Temp. Conditioning	3.6.1	4.6.1	12																
Norm. Temp. Conditioning	3.6.2	4.6.2		12															
High Temp. Conditioning	3.6.3	4.6.3			12														
Environmental Tests																			

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TABLE III. Design certification inspection

			I	II	III	IV	V	VI	VII	VIII-A	VIII-B	VIII-C	VIII-D	IX	X	XI	XII	XIII	XIV
Salt Fog	3.6.6	4.6.6				12													
- Visual Inspection	3.5.1	4.5.1				12													
- Leakage	3.5.3	4.5.3				12													
Iced Condition	3.6.8	4.6.8					3												
- Visual Inspection	3.5.1	4.5.1					3												
Dust	3.6.9	4.6.9						3											
- Visual Inspection	3.5.1	4.5.1						3											
High Temperature	3.6.7	4.6.7							6										
- Visual Inspection	3.5.1	4.5.1							6										
Sequential Program																			
20 G Shock	3.6.10	4.6.10								36									
- Visual/X-ray/Leak	3.5.1,.2,.3	4.5.1,.2,.3								36									
TSHA Cycling	3.6.11	4.6.11									27								
- Visual/X-ray/Leak	3.5.1,.2,.3	4.5.1,.2,.3									27								
Low Temp. (-60°F)	3.6.12	4.6.12										18							
- Visual	3.5.1	4.5.1										18							
-X-ray/Leak (a)	3.5.2,.3	4.5.2, .3										17							
Vibration Non-Gunfire	3.6.13.1	4.6.13.1											9						
- Visual/X-ray/Leak	3.5.1,.2,.3	4.5.1,.2,.3											9						
Vibration (Gunfire)	3.6.13.3	4.6.13.3											9						
- Visual/X-ray/Leak	3.5.1,.2,.3	4.5.1,.2,.3											9						
Safety Tests																			
Six Foot Drop	3.6.4	4.6.4												3					
- Visual/X-ray/Leak	3.5.1,.2,.3	4.5.1,.2,.3												3					
Forty Foot Drop	3.6.5	4.6.5													3				
- Visual/X-ray	3.5.1,.2	4.5.1,.2													3				
Firing Mechanism	3.5.2, .3	4.5.2, .3														6			
Propellant Tests																			
Cookoff	3.6.14	4.6.14															12		
- Visual Inspection	3.5.1	4.5.1															12		
High Temperature Exposure	3.6.15	4.6.15																12	
- Visual Inspection	3.5.1	4.5.1																12	
Performance Tests																			
Function at -80°F	3.7	4.7					3												
Function at -65°F	3.7	4.7	12			12		1		3	3	3	3			2			
Function at 70°F	3.7	4.7		12				1		3	3	3	3	3		2		12	
Function at 200°F	3.7	4.7			12			1	6	3	3	3	3			2			
Spare Units																			2

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TABLE IV. Production inspection

Item	Nature of test	Requirement	Test method	Classification (DOD-STD-2101)
Primary Component	Single lot verification	3.4	4.4	(M101)
Output Material	Type and Weight	MBEU148334-1	-	(C1)
Initiator	Visual inspection	3.5.1	4.5.1	(M102)
Initiator	Radiographic examination	3.5.2	4.5.2	(M103)
Initiator	Leakage	3.5.3	4.5.3	(M104)
Initiator	Residual magnetism	3.5.4	4.5.4	(M105)
Initiator	Serial number identification	3.2	4.2	(M106)
Initiator	Firing Mechanism	3.5.5	4.5.5	(M107)

4.1.4 Lot acceptance inspection. All lot acceptance initiators shall be inspected and tested as specified in Table V. Failure of any initiator to comply with the requirements listed in Table V shall be cause for rejection of the lot represented. Packaging defects shall be corrected before acceptance.

TABLE V. Lot acceptance inspection

Nature of test	Requirement	Test method	Classification (DOD-STD-2101)
Visual inspection	3.5.1	4.5.1	(M102)
Radiographic examination	3.5.2	4.5.2	(M103)
Leakage	3.5.3	4.5.3	(M104)
Residual magnetism	3.5.4	4.5.4	(M105)
Ballistic performance	3.7	4.7	(C2)
Preservation-packaging, packing and marking	5.0	4.5.1	(M108)

4.1.4.1 Lot sample selection. Unless otherwise specified in the contract, the initiator lot acceptance test sample size shall be in accordance with Table VI. This sample shall be randomly selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. The initiator sample size shall contain the test and retained sample quantity from the production lot.

TABLE VI. Lot sample selection

Lot Size	Production Install Size	Test Sample Size	Firing Mechanism Size	Retained Sample Size
13 to 61	2 to 50	8	5	3
67 to 106	51 to 90	13	5	3
114 to 173	91 to 150	20	5	3
176 to 305	151 to 280	32	5	3
334 to 553	281 to 500	50	5	3

4.1.4.2 Packing, packaging, and marking. The inspector shall ascertain that the packing and packaging of the initiators and the container marking conform to Section 5 of this specification and the contract.

4.2 Serial number. Verification shall occur by inspection.

4.3 Flight critical parts. Verification shall occur by inspection and documentation review.

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4.4 Design and construction. Design and construction shall be verified by inspection of the drawings and ammunition data cards, and by prototype demonstration.

4.5 Performance conditions.

4.5.1 Visual inspection. All initiators shall be verified by visual inspection and shall have no defects. Any initiator exhibiting such defects shall be rejected. Identification marking contract/drawing requirements shall also be verified. Lot numbers shall be in accordance with MIL-STD-1168

4.5.2 Radiographic examination. Each initiator shall be verified by radiographic examination in accordance with ASTM E 1742; any observable defects or degradation in assembly shall be cause for rejection. The initiator shall be positioned for the most revealing exposure with the long axis of each assembly perpendicular to the rays of the x-ray camera. All initiators shall be identified with a serial number prior to examination. The initiator serial numbers shall be in consecutive order beginning with the lowest number in each production lot. The initiators shall be arranged on trays or boards in consecutive numerical order, and each radiograph shall carry a permanent identification of the initiator displayed thereon. The radiograph identification shall include the complete lot number and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the x-ray film and those defective initiators removed from the lot.

4.5.3 Leakage.

4.5.3.1 Production installs. All initiators in the production lot shall be leak tested in an appropriate dry gas leak tester. The helium bombing time shall be for 24 hours. After removal of the initiators from the bombing chamber they shall be washed in nitrogen and then set aside for 4 hours. At the conclusion of the four hours the leakage test will be performed using appropriate detection equipment.

4.5.3.2 Non-Production installs. The initiators subjected to test shall be leak tested in an appropriate dry gas leak tester. The helium bombing time shall be for 60 minutes. After removal of the initiators from the chamber they shall be washed in nitrogen and then leakage tested using the appropriate detection equipment. There shall be no wait time prior to testing.

4.5.4 Residual magnetism. Each initiator shall be verified by testing for the lack of magnetism by being moved past a pilot's standby compass, which conforms to SAE AS5604 or equivalent. The initiator shall be moved past the compass face at a distance of six inches. This procedure shall be repeated for every 90° rotation of the initiator. If the deflection is excessive (more than 3°), the unit shall be degaussed and the test repeated until the initiator passes the test.

4.5.5 Firing mechanism test. The initiators shall be verified by inspection, demonstration and test. The sample shall be divided into three groups for paragraphs 4.5.5.2 and 4.5.5.3. Initiators conditioned prior to testing at 200 °F and –65 °F shall be conditioned for a period of time greater than 4 hours but not more than 24 hours. Initiators conditioned at 70 °F shall be conditioned for a period of time greater than 4 hours. Each test shall start within 5 minutes from conditioning chamber removal.

4.5.5.1 Free movement. Free movement shall be verified by inspection (drawing review) and prototype demonstration.

4.5.5.2 Non-Actuation. A pressure of 400 +0, -25 psig air pressure shall be gradually applied at a rate approximately 200 psig/sec through the initiators inlet port. Once reaching 400 psig this pressure shall be held for 1 minute and then released.

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4.5.5.3 Actuation. Each initiator after the non-actuation test shall be submitted to the actuation test. A pressure of 650 -0,+10 psig air pressure shall be applied at a rate between 20,000-40,000 psig/sec to the inlet port.

4.6 Environmental conditions. Inspection and test shall verify the acceptability of the initiators.

4.6.1 Low temperature conditioning. The initiators shall be verified by inspection and test by conditioning for a period of time greater than 12 hours at –65 °F, but not more than 24 hours. At the end of the conditioning, each device shall be visually inspected for degradation due to the conditioning environment. Each initiator shall be ballistically tested within five minutes from removal from the conditioning chamber.

4.6.2 Normal temperature conditioning. The initiators shall be verified by inspection and test by conditioning for a period of time greater than 12 hours at 70 °F. At the end of the conditioning, each device shall be visually inspected for degradation due to the conditioning environment. Each initiator shall be ballistically tested within five minutes from removal from the conditioning chamber.

4.6.3 High temperature conditioning. The initiators shall be verified by inspection and test by conditioning for a period of time greater than 12 hours at 200 °F, but not more than 24 hours. At the end of the conditioning, each device shall be visually inspected for degradation due to the conditioning environment. Each initiator shall be ballistically tested within five minutes from removal from the conditioning chamber.

4.6.4 Six-foot drop. Inspection and test shall verify the initiators by being submitted to the six-foot drop test outlined in MIL-C-83124 with the below requirements.

a. The initiators shall be dropped without protective caps from a height of six feet onto a 2-inch thick steel plate supported by reinforced concrete. One new initiator shall be used for each drop orientation. The initiators shall be attached to a sliding guide by string or equivalent method. This guide shall properly lead the initiator to impact according to the required orientations.

Impact Orientation

Spigot Up
Spigot Down
Threads - Side

b. After each initiator is dropped, a visual inspection shall occur to document any minor or major external defects that occurred due to the drop. To verify lack of internal defects to the initiator, each initiator shall be subjected to x-ray examination and then a leakage check.

4.6.5 Forty-foot drop. The initiators without protective devices shall be verified by inspection and test by being submitted to the forty-foot drop test outlined in MIL-STD-331, Test 103 with the below requirements.

a. The initiators shall be dropped from a height of 40 feet onto a 2-inch thick steel plate supported by reinforced concrete. One new initiator shall be used for each drop orientation. The initiator shall be attached to a sliding guide by string or equivalent method. This guide shall properly lead the initiator to impact according to the required orientation.

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Impact Orientation

Spigot Up
Spigot Down
Thread – Side

b. After each initiator is dropped, a visual inspection shall occur to document any external defects that occurred. An assessment on its ability of being disposed of safely will be done at this time. If the devices did not initiate, each initiator will be submitted to a radiograph examination to determine the extent of any internal defects or degradation.

4.6.6 Salt fog. Test and inspection shall verify the initiator by being submitted to the salt fog test in accordance with the procedures in MIL-STD-810, Method 509.2. The length of the test shall be 168 hours. The initiator shall be oriented at a 20° angle from the vertical with the output port facing downward. A 4-inch minimum distance shall be maintained around each initiator. Following this test, all initiators shall be visually inspected and leakage tested. All defects and degradation shall be noted.

4.6.7 High temperature. Test and inspection shall verify the initiator by being subjected to a high temperature test per Method 501 of MIL-STD-810 except that the initiator shall be exposed to a temperature of 200 °F until thermal stabilization is reached. This temperature shall be maintained for a period of 50 hours. The initiator shall be visually inspected and then ballistically fired at 200 °F within five minutes.

4.6.8 Iced condition. Verification shall be by inspection and test. The temperature of the initiators with protective caps installed shall be reduced to –80 ± 5 °F and stabilized. The initiators shall then be transferred to an atmosphere having a temperature of approximately 100 °F and 90% relative humidity until all evidence of frost disappears. Retaining all of the condensation practical, the temperature of the initiators shall again be stabilized at –80 ± 5 °F and ballistically tested. The functional firing shall be done within three minutes of removal from the conditioning chamber. A visual inspection for defects shall occur before and after ballistic testing.

4.6.9 Dust. Verification shall be by inspection and test by being submitted to the dust environment in accordance with MIL-STD-810, Method 510.3. The initiators, with protective caps on the inlet ports, shall be suspended in the test cabinet by waxed strings, output port down at a 20° angle from the vertical. A 3-inch minimum distance shall be maintained around each initiator and the test chamber. The following steps shall be taken.

Step 1: The chamber controls shall be set to maintain an internal temperature of 73 °F and a relative humidity of less than 30%. The air velocity shall be adjusted to 1750 ± 250 ft/min. The dust feeder shall be adjusted to maintain the dust concentration at 0.3 ± 0.2 g/ft^3. These conditions shall be maintained for 6 hours.

Step 2: The dust feed shall be stopped and the air velocity reduced to 300 ± 200 ft/min. The air temperature shall be raised to 145 °F. These conditions shall be maintained for 16 hours.

Step 3: The chamber shall be held at 145 °F and the air velocity adjusted to 1750 ± 250 ft/min. The dust feeder shall be adjusted to control the dust concentration at 0.3 ± 0.2 g/ft^3. These conditions shall be maintained for 6 hours.

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Step 4: The chamber shall be turned off and allowed to return to standard ambient conditions. The accumulation of dust shall be removed from each device by brushing, wiping or gently shaking, taking care to avoid introduction of additional dust into the initiators. The dust shall not be removed by air blast or vacuum cleaning.

Following this test, all initiators shall be visually inspected. All defects and degradation shall be noted.

4.6.10 20G Shock. Verification shall be by inspection and test. The initiators shall be submitted to the operation shock test for flight vehicle equipment outlined in MIL-STD-810, method 516, procedure I with the below requirements.

a. The initiator shall be tested in test setup 30003-3205AS284 that tests four initiators at a time. The installation shall simulate the actual service mounting characteristics.

b. The shock pulse "wave" form shall be terminal peak saw-tooth. The peak amplitude shall be 20 g with duration of 11 millisecond as shown in Figure 1. Three shocks shall be applied in each direction of three mutually perpendicular axes. Each initiator shall be examined after each axis direction for evidence of mechanical failure or any other defects that may affect proper installation or functioning.

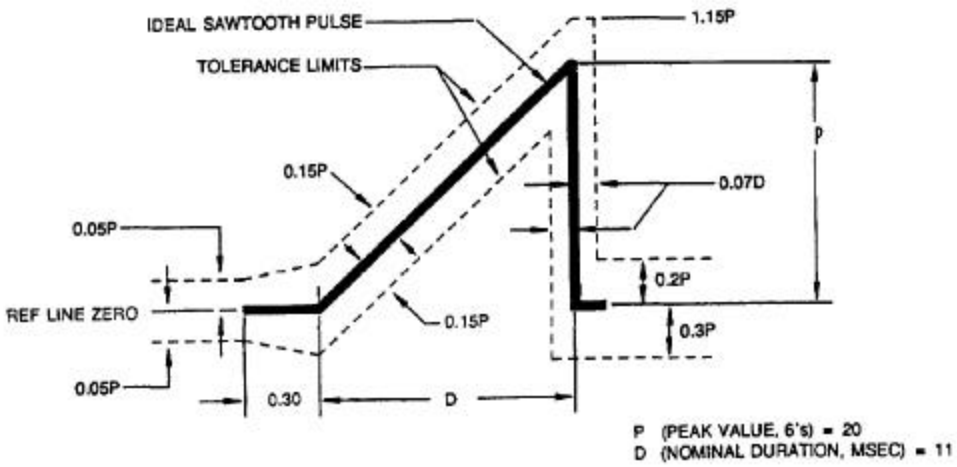


FIGURE 1. Terminal peak sawtooth shock pulse

c. After the completion of the 20-g shock test, the following performance conditions; visual inspection, radiographic examination, and leak inspection shall be conducted on each initiator to check for degradation or defects.

4.6.11 Temperature, shock, humidity, and altitude cycling. Verification shall be by inspection and test. The initiators shall be subject to the environment outlined in MIL-C-83124 with the below requirements.

a. The initiator shall be supported so the output port is pointing downward at a 20° angle from the vertical. All areas are to be exposed to the prescribed atmospheric conditions at all times throughout the test. No initiator shall be nearer than 3 inches to the chamber wall or another initiator during the test.

b. The initiators shall be submitted to the cycling schedule shown in Table VI. There are no interim withdrawals during the test.

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c. After the completion of the cycling test, the following performance conditions: visual inspection, radiographic examination, and leak inspection shall be conducted on each initiator to check for degradation and defects.

TABLE VI. Temperature shock/humidity/altitude cycling schedule

Monday:	
800	Place initiators in a chamber maintained at +70 °F at 50% relative humidity (RH).
1200	Raise chamber temperature to +160 °F and the RH to 95%. The chamber temperature shall reach +160 °F at 95% RH no later than 1300.
1600	Remove initiators from chamber and immediately place in a chamber maintained at +160°F at a pressure altitude of 70,000 feet/{0.65 psi}.
Tuesday:	
800	Remove initiators from chamber and immediately place in a chamber maintained at +70 °F at 50% RH.
1200	Remove initiators from chamber and immediately place in a chamber maintained at –65 °F at a pressure altitude of 70,000 feet/{0.65psi}.
1600	Remove initiators from chamber and immediately place in a chamber maintained at +160 °F at 95% RH.
Wednesday:	
0800	Reduce chamber temperature to +70 °F at 50% RH. The chamber temperature shall reach +70 °F conditions at 50% RH no later than 0900.
1200	Raise chamber temperature to +160 °F at 95% RH. The chamber temperature shall reach +160 °F at 95% RH no later than 1300.
1600	Remove initiators from chamber and immediately place in chamber maintained at –65 °F at a pressure altitude of 70,000 feet/{0.65 psi}.
Thursday:	
0800	Remove initiators from chamber and immediately place in a chamber maintained at +70 °F at 50% RH.
1200	Remove initiators from chamber and immediately place in a chamber maintained at –65 °F at a pressure altitude of 70,000 feet/{0.65 psi}.
1600	Remove initiators from chamber and immediately place in chamber maintained at +160 °F at 95% RH.
Friday:	
0800	Reduce chamber temperature to +70 °F at 50% RH. The chamber temperature shall reach +70 °F at 50% RH no later than 0900.
1200	Raise the chamber temperature to +160 °F at 95% RH. The chamber temperature shall reach +160 °F at 95% RH no later than 1300.
1600	Remove initiators from chamber and immediately place in a chamber maintained at –65 °F at standard ambient pressure

This schedule shall be followed for a total of four (4) weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 Monday at a temperature of +160 °F at 95% RH.

4.6.12 Low temperature. Verification shall be by inspection and test. The initiator shall be subjected to the low temperature environment outlined in MIL-STD-810, method 502.3, procedure I with the below requirements.

a. The initiators shall be temperature conditioned for 72 hours but not to exceed 74 hours at a temperature of –60 °F.

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b. After the completion of the low temperature test, the following performance conditions: visual inspection, radiographic examination, and leak inspection shall be conducted on each initiator to check for defects.

4.6.13 Vibration. Verification shall be by inspection and test. For the non-gunfire vibration test, a third of the initiators shall be tested at –65 °F, a third at 70 °F and a third at 200 °F. After completing the non-gunfire vibration testing, these same initiators shall be subjected to the gunfire vibration test at 70 °F. The initiators shall be installed into fixture 30003-3205AS284 that simulates the actual aircraft installation. The initiators shall be vibrated along each of their three mutually perpendicular axes.

4.6.13.1 Non-Gunfire vibration

a. *Resonance Search [5-2,000 Hz]:* A resonance search shall be conducted at 70 °F in the axis I orientation. The frequency sweep shall be made from 5 – 2000 Hz in approximately 10 minutes at the test levels shown in Figure 2. The resonant points shall be noted and a transmissibility plot obtained. A resonant point is defined as a point which has a transmissibility (input/output) of two or greater. If more than four resonant points are noted, only the four most severe will be used for the resonance dwell vibration. (Determining if a search is possible/applicable is dependent on the initiator mounting characteristics of the test fixture.)

b. *Resonance Dwell [5-2,000 Hz]:* The initiator will be vibrated along axis I at 70 °F for 7.5 minutes at each resonant point. The four most severe resonant points shall be used for resonance dwell. The response shall be peaked during the dwell period. The test levels and schedule are shown in Figure 2 and Table VII respectively. After resonance dwell along axis I, the initiator will be visually inspected for degradation or defects.

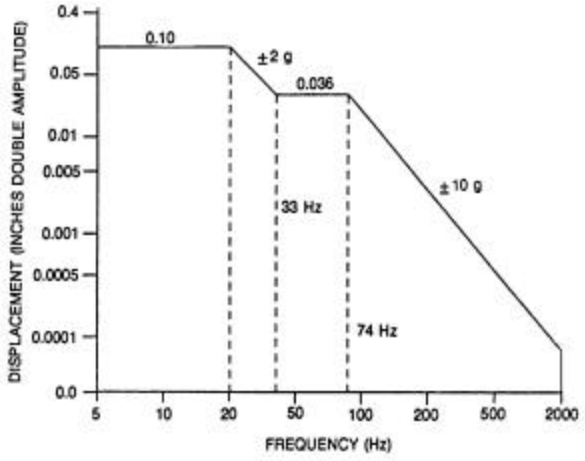


FIGURE 2. Non-Gunfire vibration test levels

c. *Sinusoidal Cycling [5-2,000 Hz] Axis I:* Each initiator shall be subjected to sinusoidal vibration cycling along axis I at 70 °F with the frequency varying logarithmically from 5–2,000 Hz and back to 5 Hz in approximately 20 minutes. The test levels and schedule are shown in Figure 2 and Table VII respectively. Depending on the number of resonant points found in the resonance search or if a search was not conducted the initiators shall be vibrated for 180 minutes minus the time spent in resonance dwell. After vibration cycling along axis I, the initiator shall be visually inspected for external degradation or defects.

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TABLE VII. Non-Gunfire sinusoidal vibration test schedule

TEST TIME PER AXIS (MINUTES)					
Number of Resonant Points	0	1	2	3	4
Dwell Time At Resonant Points	0	30	60	90	120
Total Vibration Cycling Time	180	150	120	90	60

Note: The vibration cycling sweep time from 5 Hz to 2,000 Hz and back to 5 Hz shall be 20 minutes.

d. *Axis II:* After completing the non-gunfire vibration program along axis I, the entire procedure outlined above shall be repeated along the axis II orientation.

e. *Axis III:* After completing the non-gunfire vibration program along axis II, the entire procedure outlined above shall be repeated along the axis III orientation.

f. *–65 °F Non-Gunfire Vibration:* After completing the non-gunfire vibration program at 70 °F, the procedure outlined above shall be repeated with new initiators and the temperature shall be –65 °F in lieu of 70 °F.

g. *200 °F Non-Gunfire Vibration:* After completing the non-gunfire vibration program at –65 °F, the procedure outlined above shall be repeated with new initiators and the temperature shall be 200 °F in lieu of –65 °F.

4.6.13.2 Post non-gunfire vibration test. At the completion of the non-gunfire test profile, all initiators shall be submitted to the following performance conditions: visual inspection, radiographic examination, and leak inspection to check for degradation and defect. Upon completion, the same initiators shall be submitted to gunfire vibration test profile.

4.6.13.3 Gunfire vibration

a. *Resonance Search [50-500 Hz]:* A resonance search shall be conducted on the initiator at 70 °F in the axis I orientation. The frequency sweep shall be made from 50 to 500 Hz in approximately 7.5 minutes at the test levels shown in Figure 3. The resonant points shall be noted and a transmissibility plot shall be obtained. A resonant point is defined as a point which has a transmissibility (input/output) of two or greater. If more than three resonant points are noted, only the three most severe will be used for resonance dwell vibration. (Determining if a search is possible/applicable is dependent on the initiator mounting characteristics of the test fixture.)

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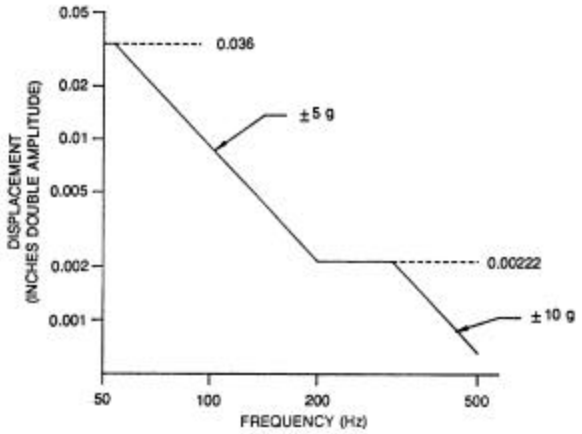


FIGURE 3. Gunfire vibration test levels

b. *Resonance Dwell [50-500 Hz]:* The initiator shall be vibrated along axis I at 70 °F for 7.5 minutes at each resonant point. The three most severe resonant points shall be used for resonance dwell. The test levels and schedule are shown in Figure 3 and Table VIII respectively. The resonance shall be peaked during the dwell period. After resonance dwell along axis I, the initiator shall be visually inspected for degradation or defects.

c. *Sinusoidal Cycling [50-500 Hz]:* The initiator shall be subjected to sinusoidal vibration cycling along axis I at 70 °F with the frequency varying logarithmically from 50 to 500 Hz and back to 50 Hz in approximately 7.5 minutes. The test levels and schedule are shown in Figure 3 and Table VIII respectively. Depending on the number of resonance points found in the resonance search or if a search was not conducted, the initiator shall be vibrated for 30 minutes minus the time spent in resonance dwell. After vibration cycling along axis I, the initiator shall be visually inspected for degradation or defects.

TABLE VIII. Gunfire sinusoidal vibration test schedule from 50 to 500 hz.

TEST TIMES IN MINUTES PER AXIS				
Number of Resonance	0	1	2	3
Sweep Time	30.00	22.50	15.00	7.50
Resonance Dwell Time	----	7.50	15.00	22.50
Fixed Dwells (See Note 1) at the following center frequencies:				
F_d = 67 Hz	7.50	7.50	7.50	7.50
100 Hz	7.50	7.50	7.50	7.50
135 Hz	7.50	7.50	7.50	7.50
200 Hz	15.00	15.00	15.00	15.00
267 Hz	7.50	7.50	7.50	7.50
300 Hz	7.50	7.50	7.50	7.50
335 Hz	7.50	7.50	7.50	7.50
400 Hz	15.00	15.00	15.00	15.00
467 Hz	7.50	7.50	7.50	7.50
500 Hz	7.50	7.50	7.50	7.50
Total Time Each Axis	120.0	120.0	120.0	120.0

[1] When equipment resonance occurs within ±5% of a fixed dwell frequency, the fixed period shall be omitted and only the resonance dwell performed. The omitted fixed dwell shall then be added to the sweep time.

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d. *Fixed Dwell Testing:* After sinusoidal cycle, the initiator shall be subjected to a series of fixed dwells at 70 °F along axis I as shown in Table VIII. The fixed narrow band dwell shall be performed sweeping the frequency about ±5% of the specified center frequency. For example, the 100 Hz. dwell should be performed from 95 to 105 Hz at the logarithmic sweep rate specified.

e. *Sinusoidal Cycling [500-2,000 Hz]:* After completing the fixed dwell testing, the initiator shall be subjected to seven sinusoidal sweeps at 70 °F along axis I. The frequency shall vary logarithmically from 500 to 2,000 Hz and return to 500 Hz in approximately 7.5 minutes with an applied input of ±10g. After testing along axis I, the initiators shall be visually inspected for defects.

f. *Axis II:* After completing the gunfire vibration program along axis I, the procedure outlined above shall be repeated along the axis II orientation.

g. *Axis III:* After completing the gunfire vibration program along axis II, the procedure outlined above shall be repeated along the axis III orientation.

4.6.13.4 Post gunfire vibration test: At the completion of the gunfire test profile, all initiators shall be submitted to the following performance conditions; visual inspection, radiographic examination, and leak inspection to check for degradation and defect.

Functional firings for the initiators shall occur at the test temperature each initiator was submitted to during the non-gunfire vibration test profile.

4.6.14 Cook-off. Verification shall occur by test and inspection. The following procedure shall be performed to determine the highest temperature at which no initiator cooks-off within a one-hour period.

a. Place 3 initiators in a temperature chamber preconditioned to 300 ± 5 °F (or temperature based on propellant information) for a period of 1 hour +5/-0 minutes.

b. Remove the initiators from the temperature chamber and visually verify the status of each initiator (functioned or not).

c. If in step b any initiators functioned, decrease the chamber temperature 25 °F. Place 3 new initiators in the temperature chamber for a period of 1-hour +5/-0 minutes.

d. Perform steps b and c until no initiator functions.

e. If in step b no initiators functioned, increase the chamber temperature 25 °F. Place 3 new initiators in the temperature chamber for a period of 1-hour +5/-0 minutes.

f. Perform step e until at least one of the initiators function.

The goal is to establish the transition temperature between functioning and non-functioning of the initiator. This transition will be the cook-off temperature.

4.6.15 High temperature exposure. Verification shall occur by test and inspection. The following procedure shall be performed to determine the highest temperature at which the initiator may be exposed for a 12-hour period and still meet ballistic conditions.

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a. Place 3 initiators in a temperature chamber preconditioned to 25 °F less than that determined for Cook-off (para 4.6.14), for a period of 12 hours +5/-0 minutes.

b. Remove the initiators from the temperature chamber and visually verify the status of each initiator (functioned or not).

c. If any initiator functioned, step e shall be performed until no initiator functions in the temperature chamber.

d. Allow the initiators to cool to 70°F. Ballistically test the initiators. If any initiators fail to meet all of the ballistic conditions, repeat steps e and d until the 3 initiators meet the ballistic parameters.

e. Decrease the chamber temperature 25 °F. Place 3 new initiators in the temperature chamber for a period of 12 hours +5/-0 minutes.

4.7 Ballistic conditions. The initiators shall be verified by test. Each initiator shall be conditioned for not less than 4 hours and no more than 24 hours at the specified temperatures. The initiators shall be removed from the conditioning chamber and test fired in conjunction with the Impulse, Cartridge, CCU-146/A within 5 minutes. If any initiator is not fired within 5 minutes after removal, it shall be reconditioned at the specified temperature for an additional 1 hour and then re-tested. The gas pressure of 900 +25, -0 psig shall be applied to the inlet port at a rate of 10,000 to 40,000 psig/sec. The test fixture temperature shall be maintained at 70 ± 10 °F prior to each ballistic firing. The initiator shall be tested in the performance firing test fixture given in drawing MBEU148480-1.

4.7.1 Integrity. The ballistically tested initiator shall be verified by inspection for integrity. A marginality of success analysis shall occur by performing a disassembly, sectioning, or other method to confirm every component within the design has maintained the proper integrity.

4.7.2 Misfires. Verification shall occur by inspection of the initiator and documentation.

4.7.2.1 Retest. Verification shall occur by documentation/data inspection.

4.7.2.2 Test failure. Verification shall occur by inspection of the initiator and test data. If a test failure is attributable to a specific cause, excluding the initiator, the original individual test results should be discarded and the complete test conducted again.

4.7.2.3 Acceptance criteria. Verification shall occur by inspection of the test results from para. 4.5 through 4.7.

4.8 Insensitive munitions. Verification shall occur by analysis or testing of the initiator in accordance with NAVSEAINST 8010.5

4.9 Explosive hazard classification. Verification shall occur by analysis or testing of the initiator in accordance with NAVSEAINST 8020.8.

4.10 Workmanship. Construction and finishing of initiators shall be verified by inspection.

5. PACKAGING

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5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (See 6.2 and 6.4). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Departments or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The ballistic gas generated from the Initiator, Cartridge Actuated, JAU-74/A provides initial pressure to the NACES P3I catapult to provide the upward movement of the ejection seat prior to catapult separation and underseat rocket motor ignition. This initiator was designed for military use and has no commercial applications.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (See 2.2.1 and 2.2.2)

c. Designation of Initiator, Cartridge Actuated (JAU-74/A)

d. When first article inspection is required (see 3.1) and, if so, specify the test activity.

e. Production lot sizes and test activity.

f. Items of data required for each inspection. (See 6.3)

g. Packaging (inner unit pack) and marking (See 5.1 and 6.4)

h. That the safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition Ashore", OP 5 are applicable.

6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a), subparagraphs below, (b) applicable documents listed in section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government unless specified in the contract or order.

6.3.1 Standard deliverables. In addition to the delivery of initiators, the contract may require delivery of the following:

a. Radiographic review report
b. Radiographic LAT sample plates and technique

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c. Requests for waiver/deviation and engineering change proposal (ECP)
d. Progress and/or management reports
e. Test and inspection report if contractor tested
f. Ammunition data card
g. Performance oriented packaging (POP) test report
h. Process control charts.
i. Department of Transportation Hazard Classification.
j. Production baseline report.

6.4 Contract packaging and marking. The following packaging and marking requirements should be specified in the contract.

6.4.1 Preservation and packing. Level A and commercial preservation/packing should be in accordance with the Code of Federal Regulation 49 CFR 171-179. The contractor is to be responsible for performance oriented packing certification and documentation in accordance with Regulation 49 CFR 171-179.

6.4.1.1 Packaging for service use. Level A preservation and packing should be used for packaging of all production lots for service use, design certification, first article and production lot acceptance inspections. Initiators should be packaged one to every hermetically sealed inner container. The number of inner containers and the method used to pack the outer container should be as approved based on the result of the accepted hazard classification testing. For an explanation of terms see 6.4.2.

6.4.2 Explanation of terms. For the purpose of this specification, NAVSUPINST 4030.28 (20 June 1975) paragraphs 1-4, 2-2, 203, and Appendix A, when applicable, is used to clarify terms.

6.4.3 Markings.

6.4.3.1 Special markings. Marking of exterior containers should be in accordance with 49 CFR 171-179. The contractor or the agent should apply markings.

6.4.3.2 Normal markings. The contract or order should specify the marking information on inner pack and the outer shipping container. The specified marking information should be applied to the containers in accordance with the applicable provisions of MIL-STD-129. Lot number should be in accordance with MIL-STD-1168.

6.5 Definition.

Ballistic parameters

a. Maximum pressure – The peak ballistic pressure reached within the test chamber.

b. Time to maximum pressure – The time interval from the first rise on the pressure curve to the peak ballistic pressure.

c. Time to ½ maximum pressure decrement – The time interval from the first rise on the pressure curve to ½ the maximum pressures decrement.

d. Ignition delay – The time interval from the first firing mechanism (i.e. shear pin failure) movement of event marker to the first rise on the output ballistic pressure curve.

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Flight critical part. A part, the single failure of which, during the operating condition, could cause loss of aircraft or one of its major components, loss of control, unintentional release of, or inability to release, or which may cause significant personnel injury, including in flight, escape, survival, or rescue.

Analysis. Verification by technical or mathematical models or simulation, algorithms, charts, graphs, or circuit diagrams, and representative data.

Defect. For the purpose of inspection or verification, the term defect is defined to mean burrs, dents, scratches, split or cracked edges, sharp edges, and all other defects which may prevent the installation, adversely affect the ballistic performance, or in any way adversely affect fit, form, function, safety or performance of the initiator.

Degrade. For the purpose of inspection or verification, the term degrade (degradation) is defined to mean chemical decomposition, fracture, yielding, debonding, delaminating, cracking, corrosion or any other condition which could adversely affect the form, fit, function or safety of this component

Demonstration. Verification by operation, adjustment, or configuration of items performing their designed functions under specific conditions. The items may be instrumented and quantitative limits or performance monitored, but only check sheets rather than actual data are required to be recorded.

Inspection. Verification by visual inspection of the item, reviewing descriptive documentation, and comparing the appropriate characteristics with predetermined standards to determine conformance to requirements without the use of special laboratory equipment or procedures.

Misfire. Failure to meet any ballistic performance condition shall be deemed a misfire.

6.6 Subject term (keyword) listing.

NACES P3I
CAD
Aircrew Escape

Preparing activity:
Navy - OS
(Project 1377-0181)

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8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial: 301-744-1973 (2) DSN: 354-1973
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